



TD3 Programmation mobile sous Android

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Exercice 1 :

On veut mettre en place une petite application Android qui permet d'afficher une liste d'utilisateurs stockés dans une base de données RoomDatabase en utilisant une RecyclerView, les détails d'un utilisateur incluent son nom, son prénom, son mail et sa ville.

Le code :

MainActivity.java :

```
package tn.rnu.enetcom.td3_ex1;

import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;
import tn.rnu.enetcom.td3_ex1.R;

public class MainActivity extends AppCompatActivity implements
View.OnClickListener {

    private Button btnAdd, btnView, btnDelete, btnUpdate;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Initialize buttons
        btnAdd = findViewById(R.id.btnAdd);
        btnView = findViewById(R.id.btnView);
        btnDelete = findViewById(R.id.btnDelete);
        btnUpdate = findViewById(R.id.btnUpdate);

        // Set click listeners
        btnAdd.setOnClickListener(this);
        btnView.setOnClickListener(this);
        btnDelete.setOnClickListener(this);
        btnUpdate.setOnClickListener(this);
    }

    @Override
    public void onClick(View v) {
        int id = v.getId();
        if (id == R.id.btnAdd) {
            startActivity(new Intent(this, AddDataActivity.class));
        } else if (id == R.id.btnView) {
            startActivity(new Intent(this, ReadDataActivity.class));
        } else if (id == R.id.btnDelete) {
            startActivity(new Intent(this, DeleteDataActivity.class));
        } else if (id == R.id.btnUpdate) {
            startActivity(new Intent(this, UpdateDataActivity.class));
        }
    }
}
```

AppDatabase.java :

```
package tn.rnu.enetcom.td3_ex1;

import androidx.room.Database;
import androidx.room.RoomDatabase;

@Database(entities = {User.class}, version = 1)
public abstract class AppDatabase extends RoomDatabase {
    public abstract MyDao myDao();
}
```

DatabaseClient.java :

```
package tn.rnu.enetcom.td3_ex1;

import android.content.Context;
import androidx.room.Room;

public class DatabaseClient {
    private Context mCtx;
    private static DatabaseClient mInstance;
    private AppDatabase appDatabase;

    private DatabaseClient(Context mCtx) {
        this.mCtx = mCtx;
        appDatabase = Room.databaseBuilder(mCtx, AppDatabase.class,
"UserDB").build();
    }

    public static synchronized DatabaseClient getInstance(Context mCtx) {
        if (mInstance == null) {
            mInstance = new DatabaseClient(mCtx);
        }
        return mInstance;
    }

    public AppDatabase getAppDatabase() {
        return appDatabase;
    }
}
```

MyAdapter.java :

```
package tn.rnu.enetcom.td3_ex1;

import android.content.Context;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;

import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;

import java.util.List;
```

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```
public class MyAdapter extends RecyclerView.Adapter<MyAdapter.ViewHolder> {
    private List<User> userList;
    private Context context;

    public MyAdapter(Context context, List<User> userList) {
        this.context = context;
        this.userList = userList;
    }

    @NonNull
    @Override
    public ViewHolder onCreateViewHolder(@NonNull ViewGroup parent, int
viewType) {
        View view = LayoutInflater.from(context).inflate(R.layout.user_row,
parent, false);
        return new ViewHolder(view);
    }

    @Override
    public void onBindViewHolder(@NonNull ViewHolder holder, int position)
{
        User user = userList.get(position);
        holder.txtid.setText(String.valueOf(user.getId()));
        holder.txtname.setText(user.getNom());
        holder.txtemail.setText(user.getMail());
        holder.txtcity.setText(user.getVille());
    }

    @Override
    public int getItemCount() {
        return userList.size();
    }

    public static class ViewHolder extends RecyclerView.ViewHolder {
        TextView txtid, txtname, txtemail, txtcity;

        public ViewHolder(@NonNull View itemView) {
            super(itemView);
            txtid = itemView.findViewById(R.id.txtId);
            txtname = itemView.findViewById(R.id.txtName);
            txtemail = itemView.findViewById(R.id.txtEmail);
            txtcity = itemView.findViewById(R.id.txtCity);
        }
    }
}
```

MyDAO.java (l'interface DAO) :

```
package tn.rnu.enetcom.td3_ex1;

import androidx.room.Dao;
import androidx.room.Delete;
import androidx.room.Insert;
import androidx.room.Query;
import androidx.room.Update;

import java.util.List;
```

```
@Dao
public interface MyDao {
    @Insert
    void insert(User user);

    @Query("SELECT * FROM User")
    List<User> getAllUsers();

    @Delete
    void delete(User user);

    @Query("DELETE FROM User WHERE id = :userId")
    void deleteById(int userId);

    @Update
    void update(User user);
}
```

AddDataActivity.java :

```
package tn.rnu.enetcom.td3_ex1;

import android.os.AsyncTask;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class AddDataActivity extends AppCompatActivity {

    private EditText editName, editEmail, editCity;
    private Button btnSave;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_add_data);

        editName = findViewById(R.id.editName);
        editEmail = findViewById(R.id.editEmail);
        editCity = findViewById(R.id.editCity);
        btnSave = findViewById(R.id.btnSave);

        btnSave.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                saveUser();
            }
        });
    }

    private void saveUser() {
        final String name = editName.getText().toString().trim();
        final String email = editEmail.getText().toString().trim();
        final String city = editCity.getText().toString().trim();
    }
}
```

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```
        if (name.isEmpty() || email.isEmpty() || city.isEmpty()) {
            Toast.makeText(this, "Please fill all fields",
                Toast.LENGTH_SHORT).show();
            return;
        }

        class SaveUser extends AsyncTask<Void, Void, Void> {

            @Override
            protected Void doInBackground(Void... voids) {
                User user = new User();
                user.setNom(name);
                user.setMail(email);
                user.setVille(city);

                DatabaseClient.getInstance(getApplicationContext())
                    .getAppDatabase()
                    .myDao()
                    .insert(user);
                return null;
            }

            @Override
            protected void onPostExecute(Void aVoid) {
                super.onPostExecute(aVoid);
                Toast.makeText(getApplicationContext(), "User added",
                    Toast.LENGTH_SHORT).show();
                finish();
            }
        }

        SaveUser saveUser = new SaveUser();
        saveUser.execute();
    }
}
```

[DeleteDataActivity.java :](#)

```
package tn.rnu.enetcom.td3_ex1;

import android.os.AsyncTask;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class DeleteDataActivity extends AppCompatActivity {

    private EditText editUserId;
    private Button btnDelete;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_delete_data);
    }
}
```

```
editUserId = findViewById(R.id.editUserId);
btnDelete = findViewById(R.id.btnDelete);

btnDelete.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        deleteUser();
    }
});

private void deleteUser() {
    final int userId =
Integer.parseInt(editUserId.getText().toString().trim());

    class DeleteUser extends AsyncTask<Void, Void, Void> {

        @Override
        protected Void doInBackground(Void... voids) {
            DatabaseClient.getInstance(getApplicationContext())
                .getAppDatabase()
                .myDao()
                .deleteById(userId);
            return null;
        }

        @Override
        protected void onPostExecute(Void aVoid) {
            super.onPostExecute(aVoid);
            Toast.makeText(getApplicationContext(), "User deleted",
Toast.LENGTH_SHORT).show();
            finish();
        }
    }

    DeleteUser deleteUser = new DeleteUser();
    deleteUser.execute();
}
```

[ReadDataActivity.java :](#)

```
package tn.rnu.enetcom.td3_ex1;

import android.os.AsyncTask;
import android.os.Bundle;

import androidx.appcompat.app.AppCompatActivity;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;

import java.util.ArrayList;
import java.util.List;

public class ReadDataActivity extends AppCompatActivity {

    private RecyclerView recyclerView;
    private MyAdapter adapter;
```

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```
private List<User> userList;

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_read_data);

    recyclerView = findViewById(R.id.recyclerView);
    recyclerView.setLayoutManager(new LinearLayoutManager(this));
    userList = new ArrayList<>();

    // Fetch data
    getData();
}

private void getData() {
    class GetData extends AsyncTask<Void, Void, List<User>> {

        @Override
        protected List<User> doInBackground(Void... voids) {
            return DatabaseClient
                .getInstance(getApplicationContext())
                .getAppDatabase()
                .myDao()
                .getAllUsers();
        }

        @Override
        protected void onPostExecute(List<User> users) {
            super.onPostExecute(users);
            userList.addAll(users);
            adapter = new MyAdapter(ReadDataActivity.this, userList);
            recyclerView.setAdapter(adapter);
        }
    }

    GetData getData = new GetData();
    getData.execute();
}
```

[UpdateDataActivity.java :](#)

```
package tn.rnu.enetcom.td3_ex1;
import android.os.AsyncTask;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
public class UpdateDataActivity extends AppCompatActivity {
    private EditText editUserId, editName, editEmail, editCity;
    private Button btnUpdate;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_update_data);
```


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```
editUserId = findViewById(R.id.editUserId);
editName = findViewById(R.id.editName);
editEmail = findViewById(R.id.editEmail);
editCity = findViewById(R.id.editCity);
btnUpdate = findViewById(R.id.btnUpdate);

btnUpdate.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        updateUser();
    }
});

}

private void updateUser() {
    final int userId =
Integer.parseInt(editUserId.getText().toString().trim());
    final String name = editName.getText().toString().trim();
    final String email = editEmail.getText().toString().trim();
    final String city = editCity.getText().toString().trim();

    if (name.isEmpty() || email.isEmpty() || city.isEmpty()) {
        Toast.makeText(this, "Please fill all fields",
Toast.LENGTH_SHORT).show();
        return;
    }

    class UpdateUser extends AsyncTask<Void, Void, Void> {

        @Override
        protected Void doInBackground(Void... voids) {
            User user = new User();
            user.setId(userId);
            user.setNom(name);
            user.setMail(email);
            user.setVille(city);

            DatabaseClient.getInstance(getApplicationContext())
                .getAppDatabase()
                .myDao()
                .update(user);

            return null;
        }

        @Override
        protected void onPostExecute(Void aVoid) {
            super.onPostExecute(aVoid);
            Toast.makeText(getApplicationContext(), "User updated",
Toast.LENGTH_SHORT).show();
            finish();
        }
    }

    UpdateUser updateUser = new UpdateUser();
    updateUser.execute();
}
```

User.java

```
package tn.rnu.enetcom.td3_ex1;

import androidx.room.Entity;
import androidx.room.PrimaryKey;

@Entity
public class User {
    @PrimaryKey(autoGenerate = true)
    private int id;

    private String nom;
    private String prenom;
    private String mail;
    private String ville;

    // Getters et Setters
    public int getId() { return id; }
    public void setId(int id) { this.id = id; }
    public String getNom() { return nom; }
    public void setNom(String nom) { this.nom = nom; }
    public String getPrenom() { return prenom; }
    public void setPrenom(String prenom) { this.prenom = prenom; }
    public String getMail() { return mail; }
    public void setMail(String mail) { this.mail = mail; }
    public String getVille() { return ville; }
    public void setVille(String ville) { this.ville = ville; }
}
```

activity_main.xml :

```
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <!-- Button to navigate to AddDataActivity -->
    <Button
        android:id="@+id/btnAdd"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Add User"
        android:layout_marginBottom="8dp" />

    <!-- Button to navigate to ReadDataActivity -->
    <Button
        android:id="@+id/btnView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="View Users"
        android:layout_marginBottom="8dp" />

    <!-- Button to navigate to DeleteDataActivity -->
    <Button
        android:id="@+id/btnDelete"
```

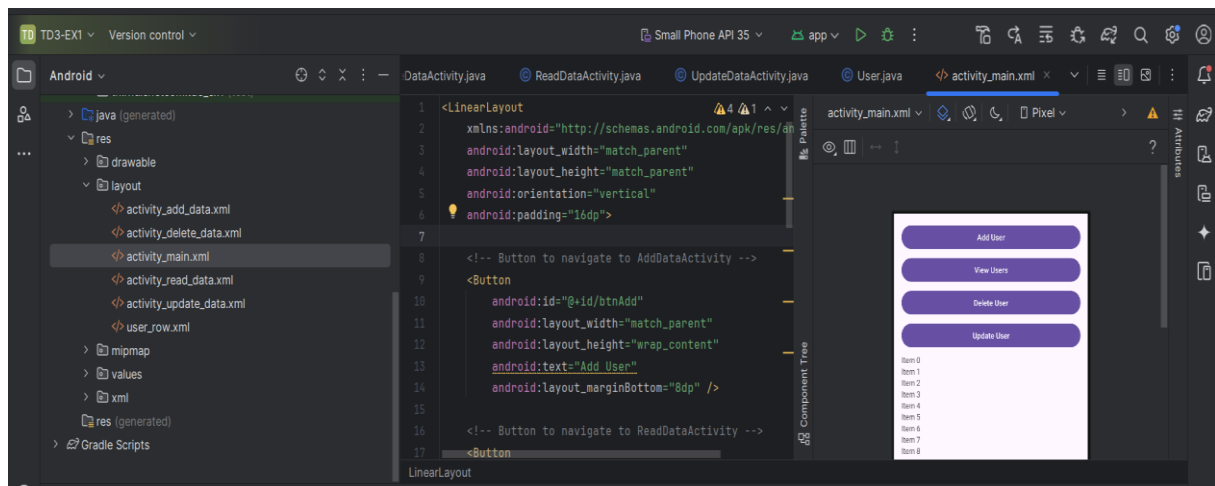
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```
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Delete User"
        android:layout_marginBottom="8dp" />

<!-- Button to navigate to UpdateDataActivity -->
<Button
    android:id="@+id/btnUpdate"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Update User"
    android:layout_marginBottom="8dp" />

<!-- RecyclerView to display the user list (optional) -->
<androidx.recyclerview.widget.RecyclerView
    android:id="@+id/recyclerView"
    android:layout_width="match_parent"
    android:layout_height="0dp"
    android:layout_weight="1" />
</LinearLayout>
```

- Vue+design :



activity_add_data.xml :

```
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <EditText
        android:id="@+id/editName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Name"/>

    <EditText
        android:id="@+id/editEmail"
        android:layout_width="match_parent"
        android:layout_height="wrap_content">
```

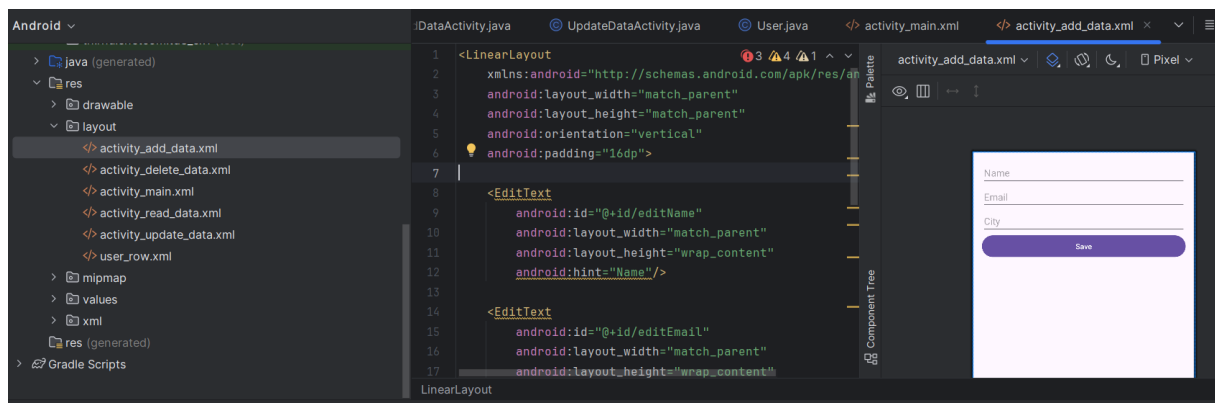
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```
        android:hint="Email"/>

        <EditText
            android:id="@+id/editCity"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:hint="City"/>

        <Button
            android:id="@+id/btnSave"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Save"/>
    </LinearLayout>
```

- Vue+design :



activity delete data.xml :

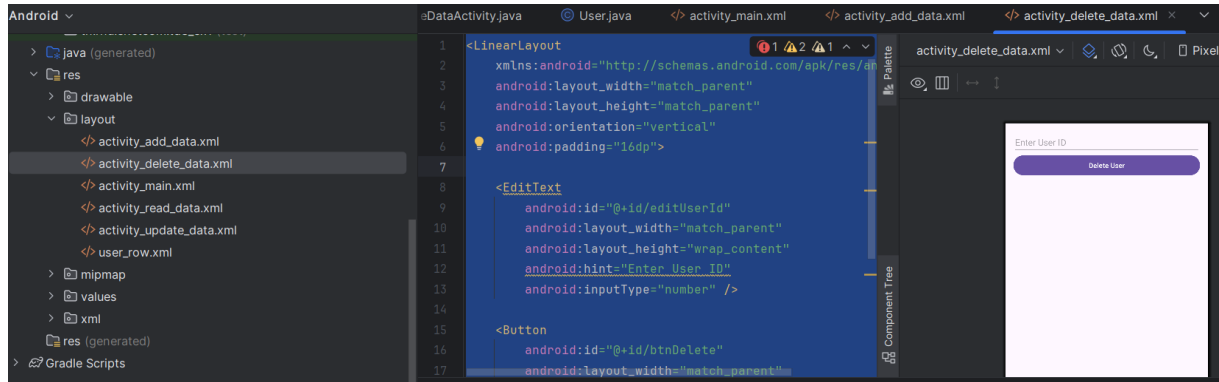
```
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <EditText
        android:id="@+id/editUserId"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter User ID"
        android:inputType="number" />

    <Button
        android:id="@+id/btnDelete"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Delete User" />
</LinearLayout>
```

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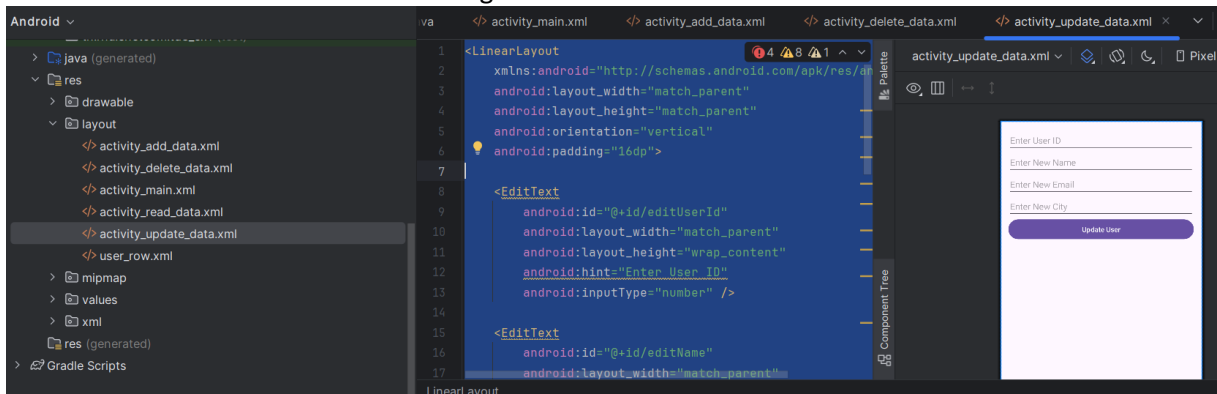
- Vue+design :



activity_update_data.xml :

```
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">
    <EditText
        android:id="@+id/editUserId"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter User ID"
        android:inputType="number" />
    <EditText
        android:id="@+id/editName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter New Name"
        android:inputType="text" />
    <EditText
        android:id="@+id/editEmail"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter New Email"
        android:inputType="textEmailAddress" />
    <EditText
        android:id="@+id/editCity"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter New City"
        android:inputType="text" />
    <Button
        android:id="@+id/btnUpdate"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Update User" />
</LinearLayout>
```

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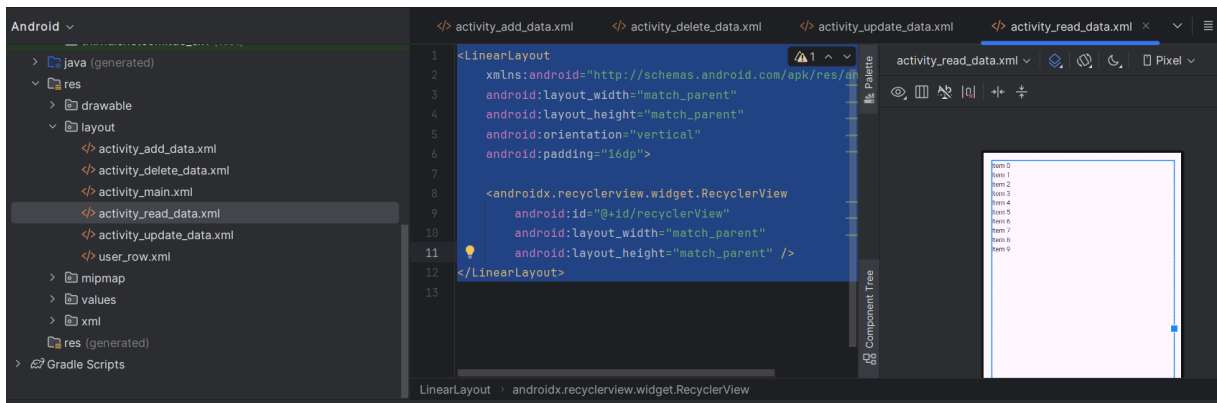
activity_read_data.xml :

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <androidx.recyclerview.widget.RecyclerView
        android:id="@+id/recyclerView"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

</LinearLayout>
```

- Vue+design :



user_row.xml :

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="8dp">

    <TextView
        android:id="@+id/txtId"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Item 0" />

    <TextView
        android:id="@+id/txtName"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Item 0" />

    <TextView
        android:id="@+id/txtEmail"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Item 0" />

    <TextView
        android:id="@+id/txtCity"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Item 0" />

</LinearLayout>
```

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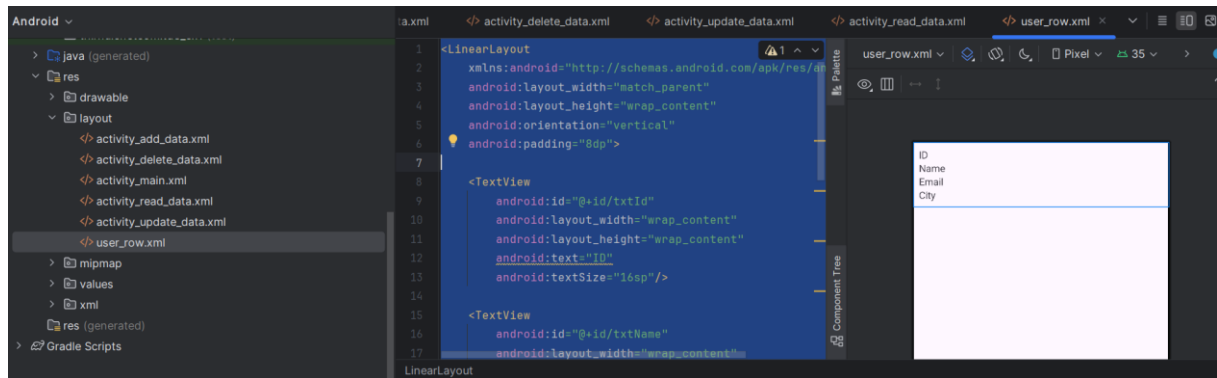
```
        android:text="ID"
        android:textSize="16sp"/>

<TextView
    android:id="@+id/txtName"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Name"
    android:textSize="16sp"/>

<TextView
    android:id="@+id/txtEmail"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Email"
    android:textSize="16sp"/>

<TextView
    android:id="@+id/txtCity"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="City"
    android:textSize="16sp"/>
</LinearLayout>
```

- Vue+design :

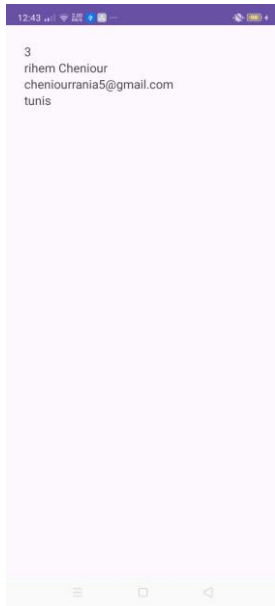


Exécution :

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The first screenshot shows a menu with four options: Add User, View Users, Delete User, and Update User. The second screenshot shows a form for adding a new user with fields for name (Rania Cheniour), email (rania.cheniour@hotmail.com), and location (sfax), followed by a Save button. The third screenshot shows a list of users with two entries: 3 (rihem Cheniour, cheniourrania5@gmail.com, tunis) and 4 (Rania Cheniour, rania.cheniour@hotmail.com, sfax).

The fourth screenshot shows a form for updating a user with fields for ID (4), name (leila), email (leila@hotmail.com), and location (tunis), followed by an Update User button. The fifth screenshot shows a list of users with two entries: 3 (rihem Cheniour, cheniourrania5@gmail.com, tunis) and 4 (leila, leila@hotmail.com, tunis). The sixth screenshot shows a numeric keypad for selecting a user ID, with the number 4 entered.



Exercice 2 :

On va reprendre l'exercice 3 du TD2 et on va ajouter une Base de données locale pour stocker la liste des tâches. Créer une application qui permet d'obtenir les interfaces suivantes, le clic sur le bouton « plus » permet d'ajouter un élément.

Le code :

MainActivity.java:

```
package tn.rnu.enetcom.ex2;

import android.content.Intent;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import androidx.recyclerview.widget.LinearLayoutManager;
import androidx.recyclerview.widget.RecyclerView;
import com.google.android.material.floatingactionbutton.FloatingActionButton;

import java.util.ArrayList;
import java.util.List;

public class MainActivity extends AppCompatActivity {

    private RecyclerView recyclerView;
    private TaskAdapter taskAdapter;
    private List<Task> taskList = new ArrayList<>();

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

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```
recyclerView = findViewById(R.id.recyclerView);
recyclerView.setLayoutManager(new LinearLayoutManager(this));
taskAdapter = new TaskAdapter(taskList);
recyclerView.setAdapter(taskAdapter);

FloatingActionButton fab = findViewById(R.id.fab);
fab.setOnClickListener(v -> {
    Intent intent = new Intent(MainActivity.this,
AddTaskActivity.class);
    startActivity(intent);
});

@Override
protected void onResume() {
    super.onResume();
    loadTasks();
}

private void loadTasks() {
    new Thread(() -> {
        List<Task> tasks =
AppDatabase.getInstance(getApplicationContext()).taskDao().getAllTasks();
        runOnUiThread(() -> taskAdapter.updateTasks(tasks));
    }).start();
}
}
```

Task.java:

```
package tn.rnu.enetcom.ex2;

import androidx.room.Entity;
import androidx.room.PrimaryKey;

@Entity
public class Task {
    @PrimaryKey(autoGenerate = true)
    private int id;

    private String title;
    private String time;

    // Getters et Setters
    public int getId() {
        return id;
    }

    public void setId(int id) {
        this.id = id;
    }

    public String getTitle() {
        return title;
    }

    public void setTitle(String title) {
```

```
        this.title = title;
    }

    public String getTime() {
        return time;
    }

    public void setTime(String time) {
        this.time = time;
    }
}
```

AppDatabase.java :

```
package tn.rnu.enetcom.ex2;

import androidx.room.Database;
import androidx.room.Room;
import androidx.room.RoomDatabase;
import android.content.Context;

@Database(entities = {Task.class}, version = 1)
public abstract class AppDatabase extends RoomDatabase {
    public abstract TaskDao taskDao();

    private static AppDatabase instance;

    public static synchronized AppDatabase getInstance(Context context) {
        if (instance == null) {
            instance =
Room.databaseBuilder(context.getApplicationContext(),
                        AppDatabase.class, "task_database")
                .fallbackToDestructiveMigration()
                .build();
        }
        return instance;
    }
}
```

AddTaskActivity.java :

```
package tn.rnu.enetcom.ex2;

import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class AddTaskActivity extends AppCompatActivity {

    private EditText editTextTitle, editTextTime;
    private Button buttonSave;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

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```
setContentView(R.layout.activity_add_task);

editTextTitle = findViewById(R.id.editTextTitle);
editTextTime = findViewById(R.id.editTextTime);
buttonSave = findViewById(R.id.buttonSave);

buttonSave.setOnClickListener(v -> saveTask());
}

private void saveTask() {
    String title = editTextTitle.getText().toString().trim();
    String time = editTextTime.getText().toString().trim();

    if (title.isEmpty() || time.isEmpty()) {
        Toast.makeText(this, "Please fill in all fields",
            Toast.LENGTH_SHORT).show();
        return;
    }

    Task task = new Task();
    task.setTitle(title);
    task.setTime(time);

    new Thread(() -> {
        AppDatabase.getInstance(getApplicationContext()).taskDao().insert(task);
        runOnUiThread(() -> {
            Toast.makeText(this, "Task added successfully",
                Toast.LENGTH_SHORT).show();
            finish();
        });
    }).start();
}
```

TaskAdapter.java :

```
package tn.rnu.enetcom.ex2;

import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;
import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;
import java.util.List;

public class TaskAdapter extends
    RecyclerView.Adapter<TaskAdapter.TaskViewHolder> {

    private List<Task> taskList;

    public TaskAdapter(List<Task> taskList) {
        this.taskList = taskList;
    }

    @NonNull
```

```
@Override
public TaskViewHolder onCreateViewHolder(@NonNull ViewGroup parent, int
viewType) {
    View view =
LayoutInflater.from(parent.getContext()).inflate(android.R.layout.simple_li
st_item_2, parent, false);
    return new TaskViewHolder(view);
}

@Override
public void onBindViewHolder(@NonNull TaskViewHolder holder, int
position) {
    Task task = taskList.get(position);
    holder.title.setText(task.getTitle());
    holder.time.setText(task.getTime());
}

@Override
public int getItemCount() {
    return taskList.size();
}

public static class TaskViewHolder extends RecyclerView.ViewHolder {
    TextView title, time;

    public TaskViewHolder(@NonNull View itemView) {
        super(itemView);
        title = itemView.findViewById(android.R.id.text1);
        time = itemView.findViewById(android.R.id.text2);
    }
}

public void updateTasks(List<Task> newTasks) {
    taskList = newTasks;
    notifyDataSetChanged();
}
}
```

TaskDao.java(interface DAO) :

```
package tn.rnu.enetcom.ex2;

import androidx.room.Dao;
import androidx.room.Insert;
import androidx.room.Query;

import java.util.List;

@Dao
public interface TaskDao {
    @Insert
    void insert(Task task);

    @Query("SELECT * FROM Task")
    List<Task> getAllTasks();
}
```

activity_add_task.xml :

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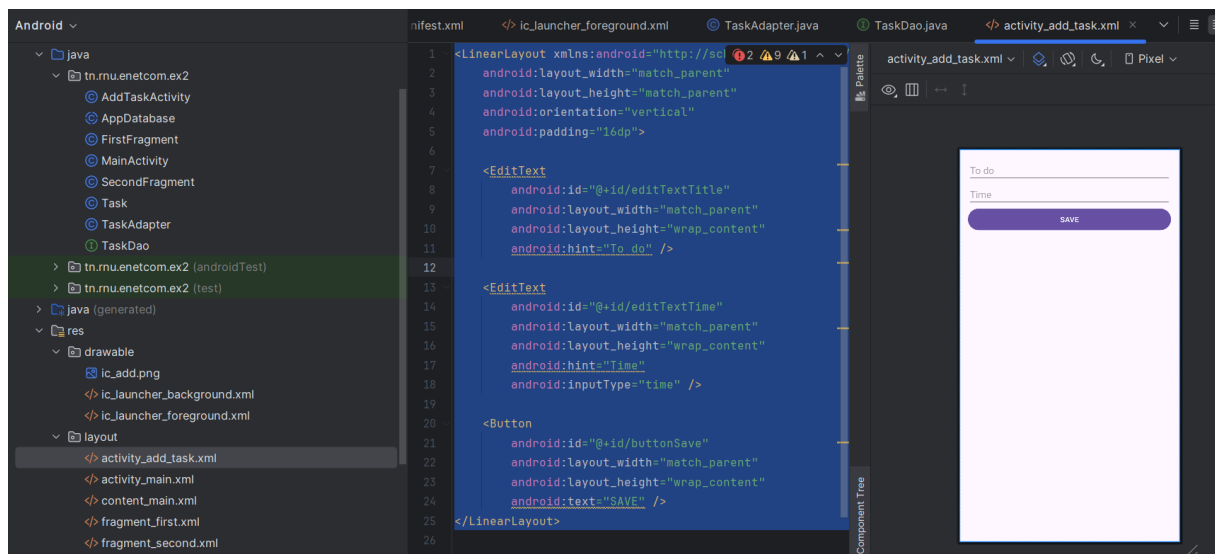
```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <EditText
        android:id="@+id/editTextTitle"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="To do" />

    <EditText
        android:id="@+id/editTextTime"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Time"
        android:inputType="time" />

    <Button
        android:id="@+id/buttonSave"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="SAVE" />

</LinearLayout>
```



Activity_main.xml :

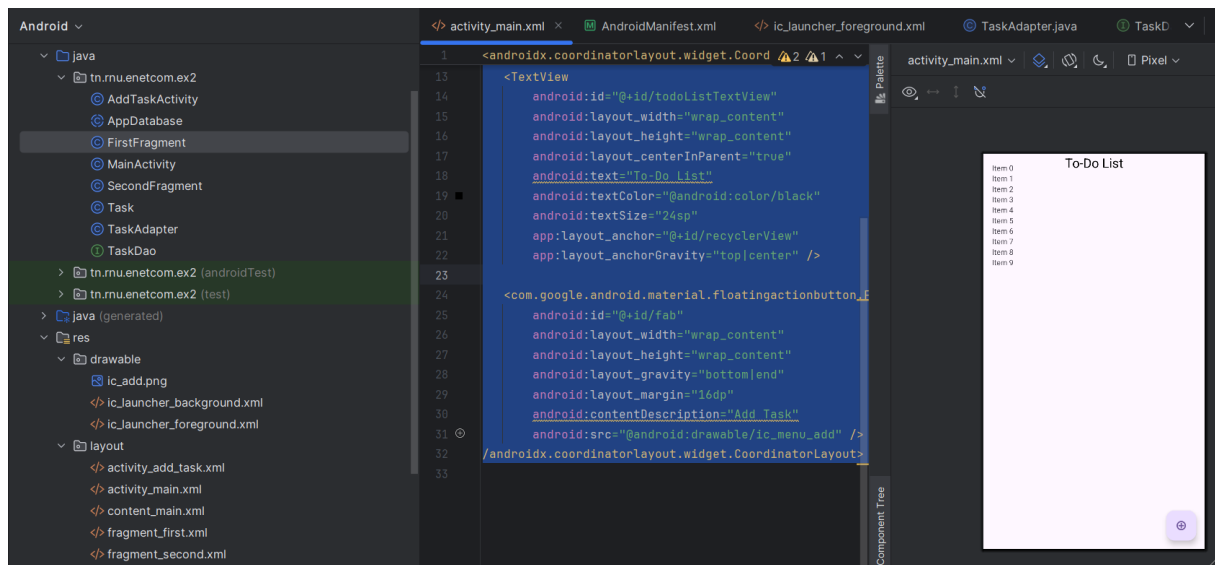
```
<androidx.coordinatorlayout.widget.CoordinatorLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <androidx.recyclerview.widget.RecyclerView
        android:id="@+id/recyclerView"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:padding="16dp" />
```

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```
<TextView
    android:id="@+id/todoListTextView"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_centerInParent="true"
    android:text="To-Do List"
    android:textColor="@android:color/black"
    android:textSize="24sp"
    app:layout_anchor="@+id/recyclerView"
    app:layout_anchorGravity="top|center" />

<com.google.android.material.floatingactionbutton.FloatingActionButton
    android:id="@+id/fab"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_gravity="bottom|end"
    android:layout_margin="16dp"
    android:contentDescription="Add Task"
    android:src="@android:drawable/ic_menu_add" />
</androidx.coordinatorlayout.widget.CoordinatorLayout>
```



L'exécution :

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